

Composite functions



1 Evaluate the following, given the functions:

$$f(x) = 4x^3$$

$$g(x) = x + 5$$

a $f(-2)$

b $g(f(-2))$

c $f(g(0))$

2 Evaluate $f(g(2))$, given the functions:

$$f(x) = 4x - 10$$

$$g(x) = 3 + \frac{3}{x}$$

3 Consider the following functions:

$$f(x) = -2x - 6$$

$$g(x) = 5x - 7$$

a Find $f(7)$.

b Hence, evaluate $g(f(7))$.

c Find $g(7)$.

d Hence, evaluate $f(g(7))$.

e Does $f(g(x)) = g(f(x))$ for all x ?

4 Find the composite function $f(g(x))$ for the following:

a $f(x) = x^3$
 $g(x) = 9x - 4$

b $f(x) = x^2$
 $g(x) = x^2 + 1$

c $f(x) = \sqrt{x}$
 $g(x) = 4x - 3$

d $f(x) = 4x + 9$
 $g(x) = x^3$

e $f(x) = 4x - 3$
 $g(x) = x^2$

5 Evaluate the following, given the functions:

$$f(x) = -2x + 2$$

$$g(x) = 4x^2 - 8$$

$$r(x) = -3x - 8$$

a $g(6)$

b $f(g(6))$

c $r(f(g(6)))$

6 Consider the following functions:



$$f(x) = -2x - 8$$

$$g(x) = 4x^2 - 4$$

- a Evaluate $g(f(6))$.
- b If $h(x)$ is defined as $f(g(x))$, state the equation for $h(x)$.
- c What type of functions are $f(g(x))$ and $g(f(x))$?

7 Consider the following functions:

$$f(x) = -2x + 6$$

$$g(x) = 3x + 1$$

- a If $r(x)$ is defined as $f(x^2)$, state the equation for $r(x)$.
- b Hence, state the equation for $q(x)$, which is $g(f(x^2))$.

8 Consider the following functions:

$$f(x) = x^2 + 3$$

$$g(x) = 4x - 9$$

- a State the equation for $f(2x)$.
- b Show that $f(2x) = g(f(x))$.

9 The function $f(x)$ is defined as $f(x) = -3x + 4g(x)$.

- a Given that $f(x)$ is a quadratic function, what type of function is $g(x)$?
- b If $f(x) = -3x + 20x^2$, find the equation for $g(x)$.
- c Find an algebraic expression for the function $g(f(x))$.

10 Consider the following functions:

$$p(x) = x + 3$$

$$q(x) = x^2 - 1$$

$$r(x) = (x + 4)(x + 2)$$

- a Rewrite $r(x)$ in expanded form.
- b Show that $r(x) = q(p(x))$.
- c Write an algebraic expression for $p(q(x))$.
- d Find the range of values for which $r(x) < p(q(x))$.

11 Consider the following functions:



$$f(x) = \frac{4}{x}$$
$$g(x) = x^2 + 3$$

Find the following functions:

a $f(x)g(x)$

b $f(g(x))$

c $\frac{f(x)}{g(x)}$

d $\frac{g(x)}{f(x)}$

12 Consider the following functions:

$$f(x) = 9x + 4$$
$$g(x) = x^2 - 2x - 3$$

Find the following functions:

a $f(x)g(x)$

b $f(g(x))$

c $\frac{f(x)}{g(x)}$

d $\frac{g(x)}{f(x)}$

Domain and range

13 For each of the following functions $f(x)$ and $g(x)$:

i Find the composite function $f(g(x))$.

ii State the domain of $f(g(x))$.

iii Find the composite function $g(f(x))$.

iv State the domain of $g(f(x))$.

a $f(x) = \frac{9}{1-2x}$
 $g(x) = \frac{3}{x}$

b $f(x) = \sqrt{x+4}$
 $g(x) = x^2 - 4$

14 Consider the function $h(x) = \sqrt{1 + \sqrt{1+x}}$ and suppose that $g(x) = \sqrt{1+x}$. Find $f(x)$ such that $h(x) = f(g(x))$.

15 Consider the function $h(x) = \frac{1}{(x-2)^6}$.

a If $g(x) = x - 2$, find $f(x)$ such that $h(x) = f(g(x))$.

b If $f(x) = \frac{1}{x}$, find $g(x)$ such that $h(x) = f(g(x))$.

16 Consider the function $f(x) = 5x^2 + 4$. Define $g(x)$ such that $g(f(x)) = f(x)$ for all x .

17 For each of the following functions:



- i State the domain and range of $f(x)$.
- ii State the domain and range of $g(x)$.
- iii Hence, determine the domain and range of $f(g(x))$.

a $f(x) = x^2$
 $g(x) = x - 7$

b $f(x) = \sqrt{x}$
 $g(x) = x - 3$

c $f(x) = \sqrt{x}$
 $g(x) = 4x + 9$

d $f(x) = x^3$
 $g(x) = x + 8$

18 Consider the following functions:

$$f(x) = \sqrt{x + 5}$$
$$g(x) = 5x - 6$$

- a Find an algebraic expression for the function $f(g(x))$.
- b Express the domain of $f(g(x))$, using inequalities.
- c Find an algebraic expression for the function $g(f(x))$.
- d Express the domain of $g(f(x))$, using inequalities.

19 Consider the following functions:

$$f(x) = -2x$$
$$g(x) = \frac{x}{x - 5}$$

- a Find an algebraic expression for the function $f(g(x))$.
- b Determine the value(s) of x that are not in the domain of $f(g(x))$.
- c Find an algebraic expression for the function $g(f(x))$.
- d Determine the value(s) of x that are not in the domain of $g(f(x))$.



Applications

- 20** At sale time in a certain clothing store, all dresses are on sale for \$5 less than 75% of the original price.
- a** Write a function $g(x)$ that finds 75% of x .
 - b** Write a function $f(x)$ that finds 5 less than x .
 - c** Construct the composite function $f(g(x))$.
 - d** Hence, calculate the sale price of a dress that has an original price of \$94.
- 21** A computer manufacturer sells hard drives to a retail outlet, each at a cost of \$11 more than the manufacturing cost. The retail store then sells each hard drive to the public, charging 60% more than they paid to the manufacturer.
- a** If m represents the manufacturing cost (in dollars), find a function $A(m)$ which returns the price of buying a hard drive at the retail store.
 - b** Find the price of buying a hard drive at the store, if the manufacturing cost is \$22.50.